

DynamicMCPBench: A Trace-Grounded, Effect-Scored Benchmark for LLM Agents over Live MCP Servers

ITMO



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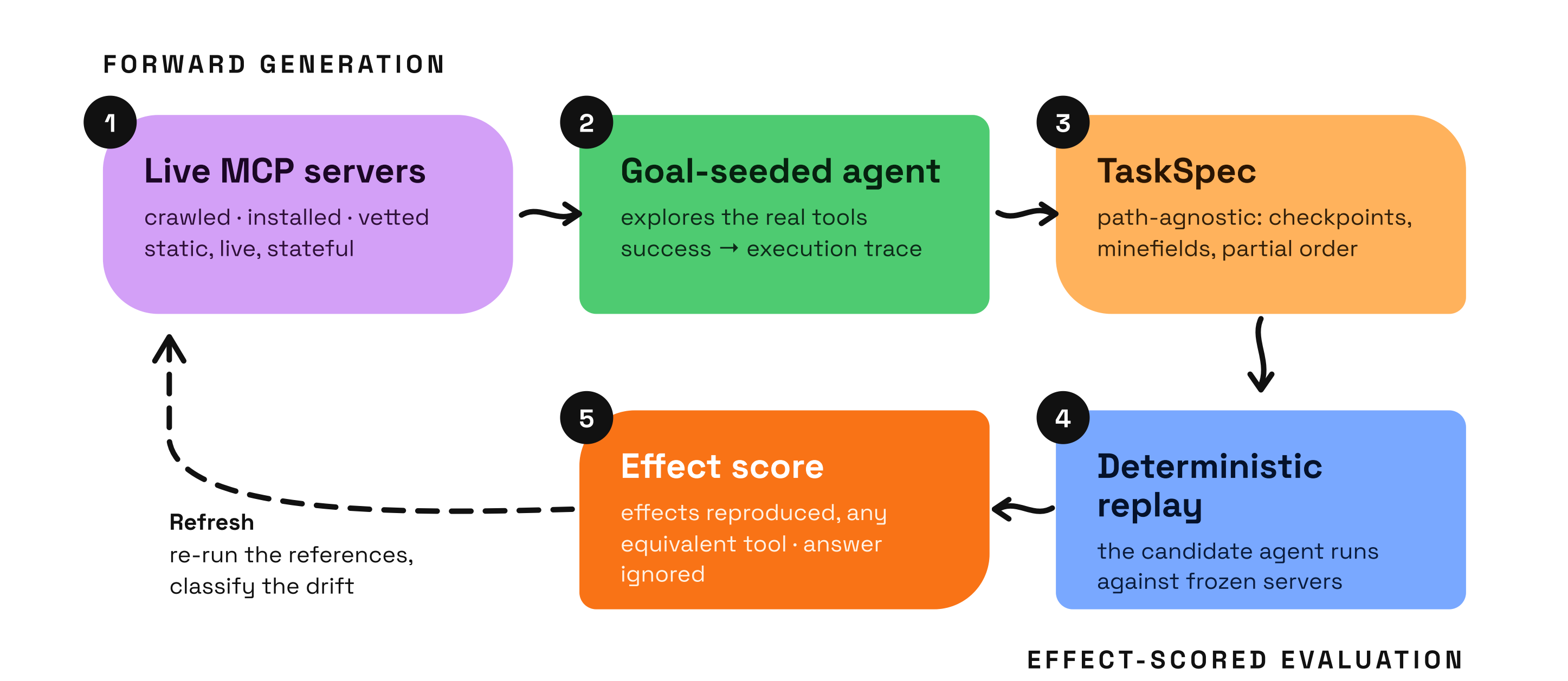
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Grade effects, not answers

Answer-matching and fixed tool lists break on live, stateful MCP servers. We record successful agent traces, distill them into path-agnostic **effect checkpoints**, and grade an agent only on whether it *reproduces those effects* — never on the answer string.

The pipeline

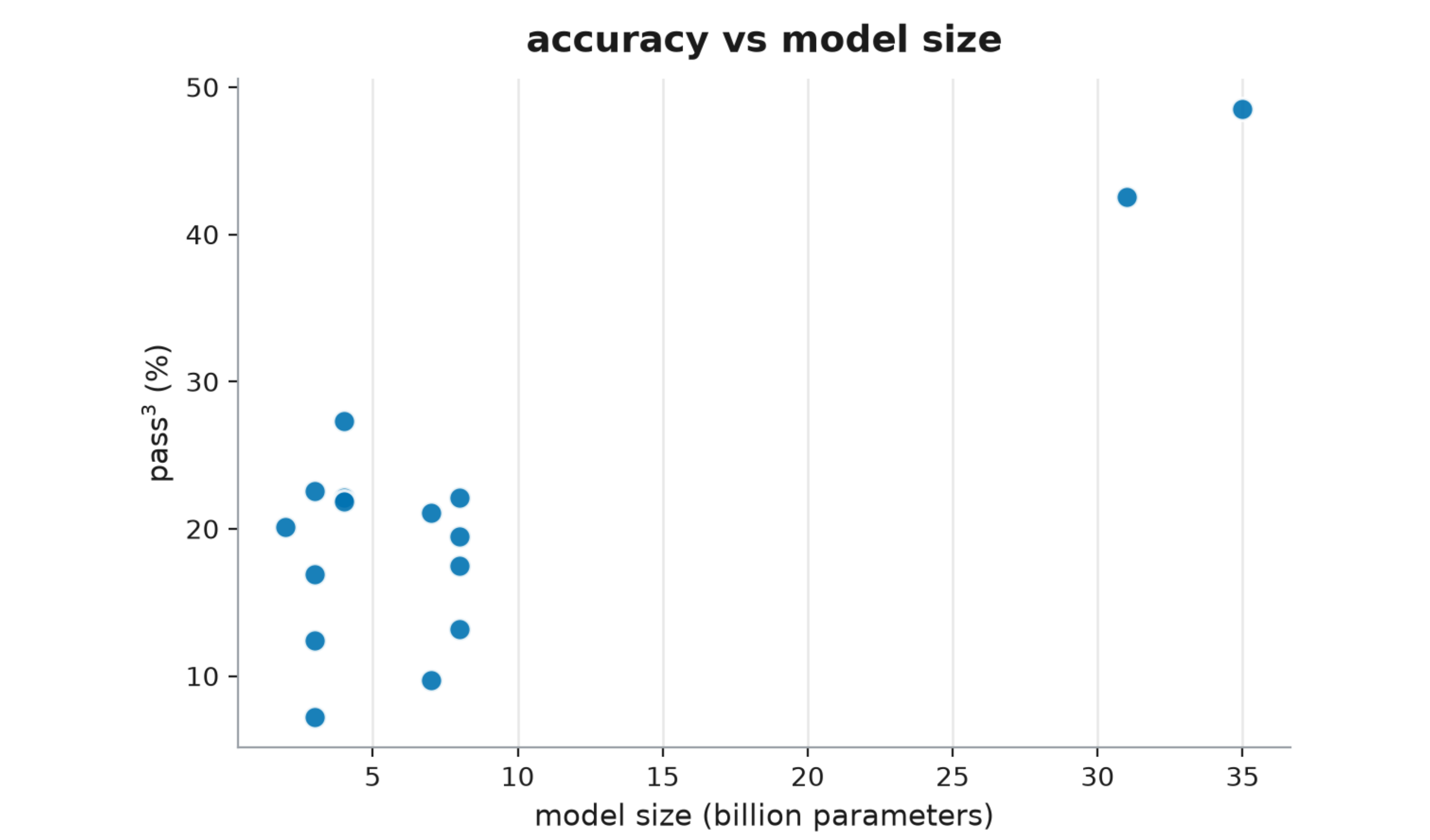


Five automated stages. (1) Crawl and vet live MCP servers, tagging each tool as live-read or stateful-write. (2) A goal-seeded agent explores the real tools and records a successful execution trace. (3) The trace is distilled into a TaskSpec: path-agnostic effect checkpoints, equivalence sets, minefields, and a partial order. (4) Candidates run under deterministic replay against frozen servers. (5) Effects are scored — any equivalent tool accepted, the answer ignored. A refresh loop re-runs the references to track drift.

How scoring works

- Effect checkpoints with **equivalence sets**: any tool that produces the required effect passes, so different valid routes all count.
- Minefields**: forbidden actions fail the run.
- Tier-1, deterministic** (no model, no network); scored **pass³** — solved only if 3/3 attempts pass.

Capability is not bought by size

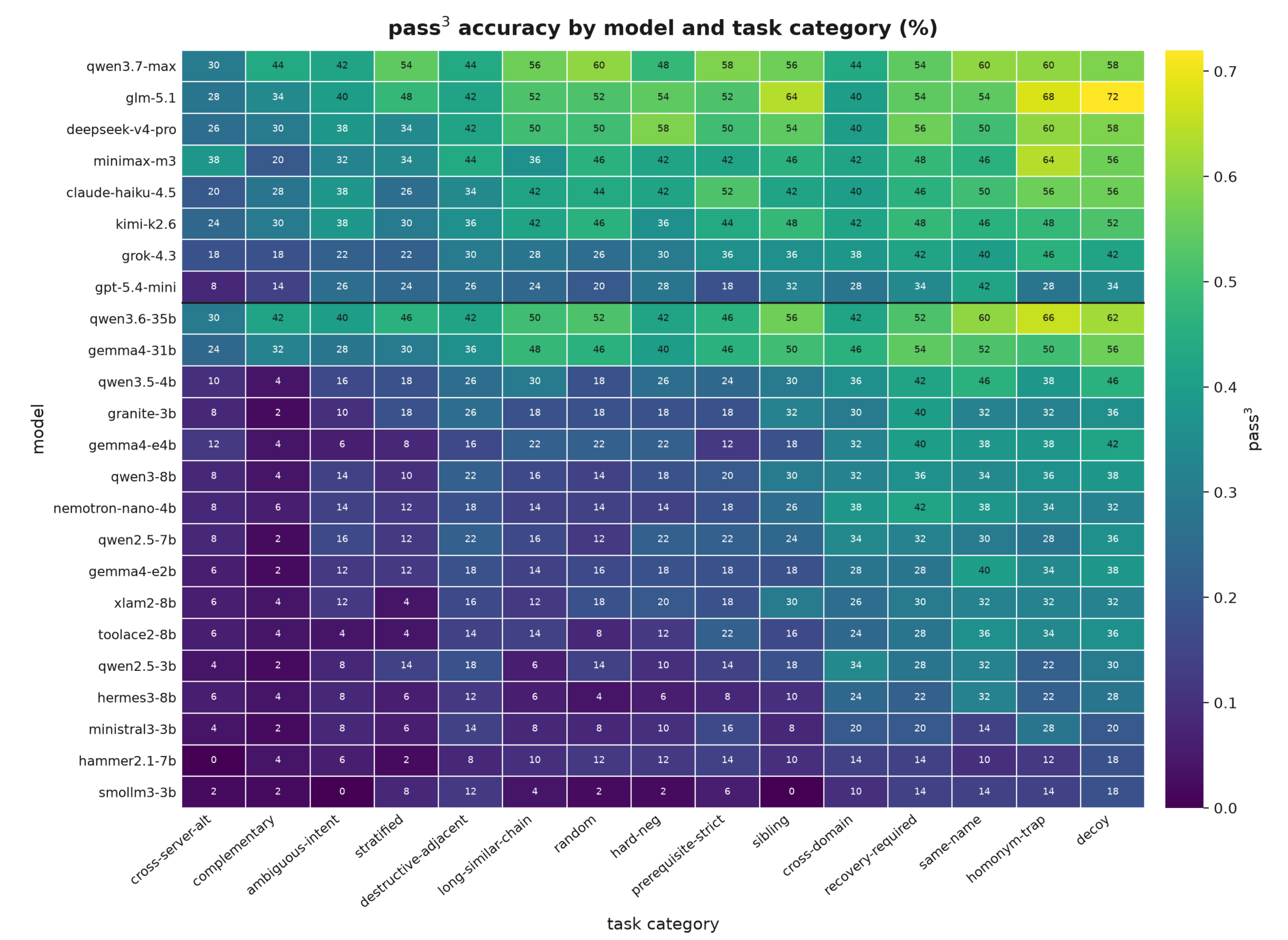


Below ~30B parameters a 4B model outperforms every 7–8B one: training and architecture matter more than raw scale.

Validated against humans

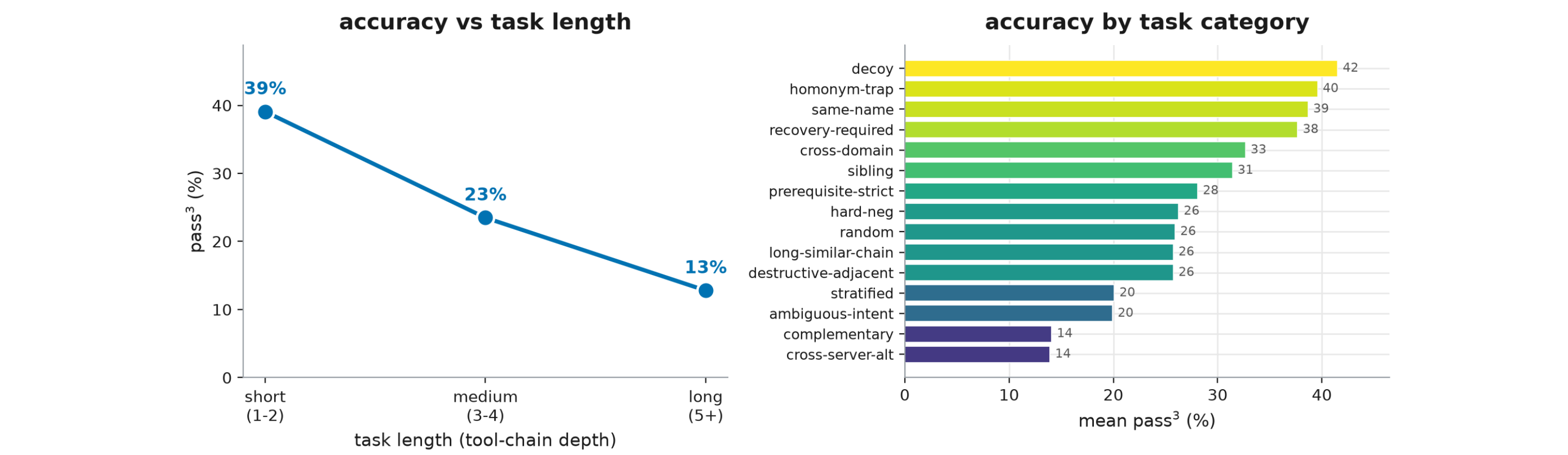
Annotators reviewed all 750 results of the strongest model: **94.9%** of automatic passes are human-confirmed, and the error is one-sided. Read every score as a **lower bound** on true capability.

The benchmark is far from solved



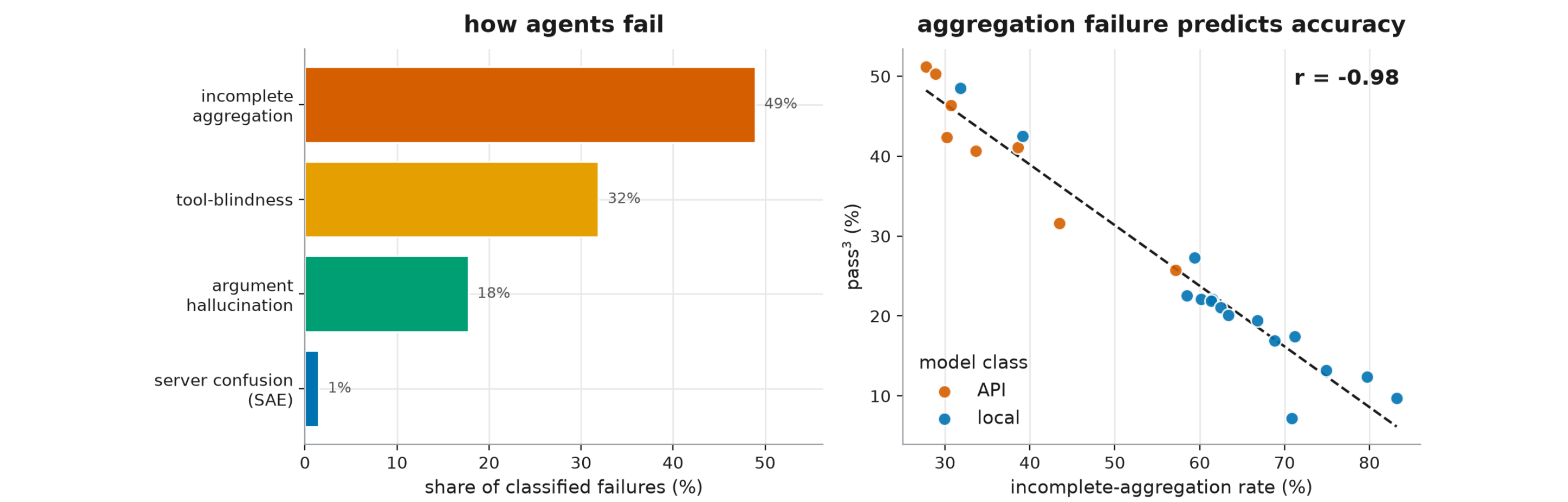
pass³ by model and category (24 models, 121 servers, 750 tasks). Difficulty is vertical — some categories are hard for every model. **31%** of tasks are solved by *no* model, only **2%** by all.

Harder as chains get longer



Accuracy falls with tool-chain length: **39%** (1–2 tools) → **23%** (3–4) → **13%** (5+). Agents fail by *not finishing* long, cross-server jobs — not by picking the wrong tool.

How agents fail



Over 54k runs, failures are dominated by **incomplete aggregation** (unmet value checkpoints) and **tool-blindness**; server confusion is near-floor. Incomplete aggregation alone predicts pass³ at $r = -0.98$.

Paper · Code · Data



Scan for the paper, the open-source framework, and the released dataset.

Run the benchmark on *your own* MCP servers.